Metadata for Brandon Parva (BRA), Norfolk, UK

Description of data set	Magnetic time-series from the BGS SWIGS differential magnetometer method (DMM) systems. Funded by NERC, grant number: NE/P017231/1 "Space Weather Impact on Ground-based Systems (SWIGS)". These data consist of measurements of the Earth's natural magnetic field at the remote site (BRAR) and the natural magnetic field plus the field created by GIC at the underline site (BRAU). The database will include .xyz files with the DMM data and one document with metadata. See Hübert et al. (2020) for further details.
Location (in lat/long): underline station (BRAU)	52.637299 N, 1.065817 E
Location (in lat/long): remote station	52.6386832 N, 1.0654730E
Description of site (i.e. open moorland)	Open arable farmland close to a boundary hedge/ditch between two fields.
Equipment used	Sensys 3DFM fluxgate magnetometer and Earth Data digitisers (EDR 209, 24bit)
Installation date	27/02/2019
Removal date	10/10/2019 (equipment reported stolen)
Number of days collected	BRAU: 22 (11-13/06/2019, 03/-16/08/2019, 28-30/09/2019, 04-05/10/2019) BRAR: 35 (06/09-10/10/2019)
Data sampling rate	1 Hz
Notes of known data gaps	Problems with the data logger at the underline site caused long gaps, see what data is available above
Qualitative estimate of site quality	Sufficient
Orientation angle of HV wires relative to true north	+6.4° (from True North)
Alignment of magnetometers	Magnetic north as of 27/02/2010
Voltage level of the lines	400 kV (Necton-Norwich)
Height above the ground of each wire (m)	Lower: 15.82 m, Middle: 25.14 m, Upper: 35.15 m
Underline magnetometer	Placed centrally, 9 m from either circuit
File Type	ASCII text (.xyz)
File format	YYYY-MM-DD HH:MM:SS X Y Z Five columns with 86400 rows (plus one header row). Magnetic values in units of nT (10^-9 T). Empty values denoted by: 999999.00
Citation	Hübert, J., Beggan, C. D., Richardson, G. S., Martyn, T., & Thomson, A. W. P. (2020). Differential magnetometer measurements of geomagnetically induced currents in a complex high voltage network. Space Weather, 18, e2019SW002421. doi: 10.1029/2019SW002421